A9

Q1

Integer Optimization Problem

Decision Variables

- x_A: Number of units of product A to produce (non-negative integer).
- x_B: Number of units of product B to produce (non-negative integer).
- x_C: Number of units of product C to produce (non-negative integer).
- y: Binary variable (0 or 1), where $y = 1$ if the total number of products exceeds 20 units, and $y = 0$ otherwise.

Objective Function

Maximize total profit, which is the profit from each product minus the storage cost if incurred:

$$\text{Maximize} \quad 30x_A + 40x_B + 50x_C - 200y$$

Constraints

1. **Machine 1 Availability**: Total hours on Machine 1 must not exceed 60 hours.
$$3x_A + 2x_B + x_C \leq 60$$
2. **Machine 2 Availability**: Total hours on Machine 2 must not exceed 72 hours.
$$2x_A + 4x_B + 3x_C \leq 72$$
3. **Minimum Production of A**: At least 8 units of A must be produced.
$$x_A \geq 8$$
4. **Minimum Production of B**: At least 10 units of B must be produced.
$$x_B \geq 10$$
5. **Minimum Production of C**: At least 5 units of C must be produced.
$$x_C \geq 5$$
6. **Total Production Limit**: Total number of products cannot exceed 40 units.
$$x_A + x_B + x_C \leq 40$$
7. **Storage Cost Condition**: If total production exceeds 20 units, $y = 1$. Modeled with a big-M constraint ($M = 20$).
$$x_A + x_B + x_C - 20 \leq 20y$$
8. **Production Allocation Constraints**: No single product can exceed 70% of total production.
   - For A:
   $$3x_A - 7x_B - 7x_C \leq 0$$
   - For B:
   $$-7x_A + 3x_B - 7x_C \leq 0$$
   - For C:
   $$-7x_A - 7x_B + 3x_C \leq 0$$
9. **Variable Domains**:
   - $x_A, x_B, x_C \geq 0$ and integer
   - $y \in \{0, 1\}$

Q2

Step 1: Solve the LP Relaxation (Problem P0)

First solve the problem without the integer constraints, i.e., $x_1, x_2 \geq 0$ are real numbers. The feasible region is defined by the constraints:

1. $2x_1 + 3x_2 \leq 12$
2. $3x_1 + 2x_2 \leq 12$
3. $x_1 \geq 0$
4. $x_2 \geq 0$

Define the Feasible Region

To find the feasible region graphically, determine the corner points by finding intercepts and intersections:

- **Constraint 1:** $2x_1 + 3x_2 = 12$
  - $x_1 = 0$: $x_2 = 4$, point (0, 4)
  - $x_2 = 0$: $x_1 = 6$, point (6, 0)
- **Constraint 2:** $3x_1 + 2x_2 = 12$
  - $x_1 = 0$: $x_2 = 6$, point (0, 6)
  - $x_2 = 0$: $x_1 = 4$, point (4, 0)
- **Intersection of Constraints 1 and 2:**
  Solve:

$$2x_1 + 3x_2 = 12 \quad (1)$$
$$3x_1 + 2x_2 = 12 \quad (2)$$

  Intersection point: (2.4, 2.4)

The feasible region is a polygon with vertices at:

- (0, 0) (origin)
- (0, 4) (Constraint 1 at $x_1 = 0$)
- (2.4, 2.4) (intersection)
- (4, 0) (Constraint 2 at $x_2 = 0$)

Maximize the Objective Function

Evaluate $3x_1 + 4x_2$ at each vertex:

- (0, 0): $3(0) + 4(0) = 0$
- (0, 4): $3(0) + 4(4) = 16$
- (4, 0): $3(4) + 4(0) = 12$
- (2.4, 2.4): $3(2.4) + 4(2.4) = 7.2 + 9.6 = 16.8$

The maximum value is 16.8 at (2.4, 2.4). Since $x_1 = 2.4$ and $x_2 = 2.4$ are not integers, now need to branch to find an integer solution. This value, 16.8, serves as an upper bound for the optimal integer solution.

Step 2: Branch on x_1

Since $x_1 = 2.4$ is between 2 and 3, branch into two subproblems:

- **P1**: $x_1 \leq 2$
- **P2**: $x_1 \geq 3$

Subproblem P1: $x_1 \leq 2$

Add the constraint $x_1 \leq 2$ to the original problem:

$$\begin{aligned} \text{maximize} \quad & 3x_1 + 4x_2 \\ \text{subject to} \quad & 2x_1 + 3x_2 \leq 12 \\ & 3x_1 + 2x_2 \leq 12 \\ & x_1 \leq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

New feasible region:

- Original vertices: (0, 0), (0, 4), (2.4, 2.4), (4, 0)
- Intersection of $x_1 = 2$ with Constraint 1: $x_2 = \frac{8}{3} \approx 2.666$
  - Point: (2, $\frac{8}{3}$)
  - Constraint 2: $3(2) + 2(\frac{8}{3}) = 6 + \frac{16}{3} = \frac{18}{3} + \frac{16}{3} = \frac{34}{3} \approx 11.33 \leq 12$, feasible

Vertices of the feasible region:

- (0, 0)
- (2, 0)
- (2, $\frac{8}{3}$)
- (0, 4) (since $2(0) + 3(4) = 12$, $3(0) + 2(4) = 8 \leq 12$)

Evaluate the objective function:

- (0, 0): 0
- (2, 0): $3(2) + 4(0) = 6$
- (0, 4): $3(0) + 4(4) = 16$
- $(2, \frac{8}{3})$: $3(2) + 4(\frac{8}{3}) = \frac{50}{3} \approx 16.667$

Maximum is $\frac{50}{3} \approx 16.667$ at $(2, \frac{8}{3})$. Since $\frac{8}{3}$ is not an integer, branch further.

Subproblem P2: $x_1 \geq 3$

Add the constraint $x_1 \geq 3$:

$$\begin{aligned} \text{maximize} \quad & 3x_1 + 4x_2 \\ \text{subject to} \quad & 2x_1 + 3x_2 \leq 12 \\ & 3x_1 + 2x_2 \leq 12 \\ & x_1 \geq 3 \\ & x_1, x_2 \geq 0 \end{aligned}$$

New feasible region:

- Original vertices: (0, 0), (0, 4), (2.4, 2.4), (4, 0)
- With $x_1 \geq 3$, exclude points where $x_1 < 3$
- $x_1 = 3$ with Constraint 2: $3(3) + 2x_2 = 12 \Rightarrow x_2 = 1.5$
  - Point: (3, 1.5)
  - Check Constraint 1: $2(3) + 3(1.5) = 10.5 \leq 12$, feasible
- Include (4, 0)

Vertices:

- (3, 0)
- (4, 0)
- (3, 1.5)

Evaluate:

- (3, 0): $3(3) + 4(0) = 9$
- (4, 0): $3(4) + 4(0) = 12$
- (3, 1.5): $3(3) + 4(1.5) = 9 + 6 = 15$

Maximum is 15 at (3, 1.5), not integer. We'll explore P2 later if needed.

Step 3: Branch on P1 ($x_1 \leq 2$)

Since P1's maximum is 16.667 > 15 (P2's maximum), and it's non-integer ($x_2 = \frac{8}{3} \approx 2.667$), branch on x_2 at 2 and 3:

- **P3**: $x_1 \leq 2$, $x_2 \leq 2$
- **P4**: $x_1 \leq 2$, $x_2 \geq 3$

Subproblem P3: $x_1 \leq 2$, $x_2 \leq 2$

$$\begin{aligned} \text{maximize} \quad & 3x_1 + 4x_2 \\ \text{subject to} \quad & 2x_1 + 3x_2 \leq 12 \\ & 3x_1 + 2x_2 \leq 12 \\ & x_1 \leq 2 \\ & x_2 \leq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Feasible region (intersection with $x_1 \leq 2$, $x_2 \leq 2$):

- Check bounds: At (2, 2), $2(2) + 3(2) = 10 \leq 12$, $3(2) + 2(2) = 10 \leq 12$, feasible
- Vertices: (0, 0), (2, 0), (2, 2), (0, 2)

Evaluate:

- (0, 0): 0
- (2, 0): 6
- (0, 2): $3(0) + 4(2) = 8$
- (2, 2): $3(2) + 4(2) = 14$

Maximum is 14 at (2, 2), an integer solution. This is the first candidate, with a lower bound of 14.

Subproblem P4: $x_1 \leq 2$, $x_2 \geq 3$

$$\begin{aligned} \text{maximize} \quad & 3x_1 + 4x_2 \\ \text{subject to} \quad & 2x_1 + 3x_2 \leq 12 \\ & 3x_1 + 2x_2 \leq 12 \\ & x_1 \leq 2 \\ & x_2 \geq 3 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Feasible region:

- $x_2 = 3$: $2x_1 + 3(3) \leq 12 \Rightarrow x_1 \leq 1.5$
- $x_1 = 0$: $3x_2 \leq 12 \Rightarrow x_2 \leq 4$
- Vertices:
  - (0, 3)
  - (1.5, 3) (since $2(1.5) + 3(3) = 12$, $3(1.5) + 2(3) = 10.5 \leq 12$)
  - (0, 4) (since $2(0) + 3(4) = 12$, $3(0) + 2(4) = 8 \leq 12$)

Evaluate:

- (0, 3): $3(0) + 4(3) = 12$
- (1.5, 3): $3(1.5) + 4(3) = 16.5$
- (0, 4): $3(0) + 4(4) = 16$

Maximum is 16.5 at (1.5, 3), not integer. Check integer points:

- $x_1 = 0$: $x_2 = 3, 4 \rightarrow$ (0, 3): 12, (0, 4): 16
- $x_1 = 1$: $x_2 = 3$: $2(1) + 3(3) = 11 \leq 12$, $3(1) + 2(3) = 9 \leq 12 \rightarrow$ (1, 3): $3 + 12 = 15$
- $x_1 = 2$: $x_2 \geq 3$: $2(2) + 3(3) = 13 > 12$, infeasible

Integer solutions: (0, 3): 12, (1, 3): 15, (0, 4): 16. Best is 16 at (0, 4), an integer solution.

Step 4: Bounding and Conclusion

- **Current Best Integer Solution**: (0, 4), value 16
- **P2**: LP maximum 15 < 16, prune (no integer solution can exceed 15)
- **P4**: LP maximum 16.5 > 16, but all integer solutions (12, 15, 16) are ≤ 16, no better solution possible
- **P0**: LP maximum 16.8 > 16, but all subproblems explored

Since 16 is the best integer solution found, and remaining subproblems cannot improve it (P2 pruned, P4 fully explored), the optimal solution is $x_1 = 0$, $x_2 = 4$, with objective value 16.

Solution: $(x_1, x_2) = (0, 4)$, objective value = 16